

JEE MAIN REVISION MASTER: SALT ANALYSIS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 43 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Salt Analysis — Applied Cases, Reactions & Traps	26 PYQs	Applied Topics
Very High	Salt Analysis — Core Principles & Governing Laws	16 PYQs	Core Principle
Standard	Salt Analysis — Standard Calculations & Problem Archetypes	1 PYQs	Quantitative Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Salt Analysis — Applied Cases, Reactions & Traps	$\text{CuSO}_4, \text{KMnO}_4, +\text{K}_4[\text{Fe}(\text{CN})_6] \rightarrow$
Salt Analysis — Core Principles & Governing Laws	$\text{NH}_4\text{OH}, x\text{Scm}^2\text{mol}^{-1}, \text{Ca}^{2+}$
Salt Analysis — Standard Calculations & Problem Archetypes	$\text{Mg}(\text{NH}_4)\text{PO}_4, \text{SrSO}_4$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Salt Analysis — Applied Cases, Reactions & Traps (Very High Yield — 26 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Salt Analysis.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2026 Morning Shift] Q#2 (MCQ): "A salt with few drops of conc. HCl gives apple green colour in flame test. The group precipitate of the salt is dissolved in acetic acid and..." [Ans: N/A]
- [2026 Morning Shift] Q#3 (MCQ): "Given below are two statements: Statement I : Griss-Ilosvay test is used for the detection of nitrite ion, which involves the use of sulphur..." [Ans: D]
- [2025 Evening Shift] Q#1 (MCQ): "Identify the inorganic sulphides that are yellow in colour : (A) $(\text{NH}_4)_2\text{S}$ (B) PbS (C) CuS (D) As_2S_3 (E) As_2S_5 Cho..." [Ans: N/A]

3.2 Salt Analysis — Core Principles & Governing Laws (Very High Yield — 16 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Salt Analysis.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Salt Analysis.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Salt Analysis PYQs.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Among Fe^{3+} , Pb^{2+} , Cu^{2+} and Mn^{2+} , identify the one that gets precipitated out while passing H_2S in presence of NH_4OH ..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Consider three metal chlorides x, y and z, where x is water soluble at room temperature, y is sparingly soluble in water at room temperature..." [Ans: N/A]
- [2026 Evening Shift] Q#4 (MCQ): "In the Group analysis of cations, Ba^{2+} & Ca^{2+} are precipitated respectively as..." [Ans: D]

3.3 Salt Analysis — Standard Calculations & Problem Archetypes (Standard Yield — 1 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Salt Analysis.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2024 Morning Shift] Q#2 (Numerical): "The number of white coloured salts, among the following is (a) SrSO_4 (b) $\text{Mg}(\text{NH}_4)\text{PO}_4$ (c) BaCrO_4 (d) $\text{Mn}(\text{OH})_2$ (e) PbSO_4 ..." [Ans: 5]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 4th April Evening Shift (Q#1)

Among Fe^{3+} , Pb^{2+} , Cu^{2+} and Mn^{2+} , identify the one that gets precipitated out while passing H_2S in presence of NH_4OH as group reagent. The highest possible oxidation state of the corresponding metal is

- (A) +3
- (B) +4
- (C) +2
- (D) +7

Correct Answer: None

Solution Analysis: While passing H_2S in the presence of NH_4OH , the medium becomes basic. In basic medium, the concentration of S^{2-} ions increases, so sulphides of some metal ions get precipitated. Now let us check the given ions: Fe^{3+} : does not precipitate here as sulphide in this group Pb^{2+} : belongs to earlier group and is precipitated as chloride, not here Cu^{2+} : precipitates as CuS Mn^{2+} : can precipitate as MnS in ammoniacal medium But in classical qualitative analysis, passing H_2S in presence of NH_4OH is used for precipitating group IV cations like Mn^{2+}

Benchmark Problem 2: JEE Main 2026 (Online) 28th January Morning Shift (Q#3)

Given below are two statements: Statement I : Griss-Ilosvay test is used for the detection of nitrite ion, which involves the use of sulphanilic acid and α -naphthylamine reagent. Statement II : In the above test, sulphanilic acid is diazotized by the acidified nitrite ion, which on further coupling with α -naphthylamine forms an azo-dye. In the light of the above statements, choose the correct answer from the options given below

- (A) Statement I is true but Statement II is false
- (B) Both Statement I and Statement II are false
- (C) Statement I is false but Statement II is true
- (D) Both Statement I and Statement II are true **[Correct]**

Correct Answer: D

Solution Analysis: In Gries-Ilosvay test for nitrite ion (NO_2^-), the reagents used are sulphanilic acid and α -naphthylamine. In acidic medium, nitrite ion forms nitrous acid HNO_2 , which diazotizes sulphanilic acid to give a diazonium salt. This diazonium salt then couples with α -naphthylamine to form a coloured azo dye (usually reddish/pink). So, Statement I is true and Statement II is also true. Correct option: D (Both Statement I and Statement II are true)

Benchmark Problem 3: JEE Main 2024 (Online) 1st February Morning Shift (Q#2)

The number of white coloured salts, among the following is (a) SrSO_4 (b) $\text{Mg}(\text{NH}_4)\text{PO}_4$ (c) BaCrO_4 (d) $\text{Mn}(\text{OH})_2$ (e) PbSO_4 (f) PbCrO_4 (g) AgBr (h) PbI_2 (i) CaC_2O_4 (j) $[\text{Fe}(\text{OH})_2(\text{CH}_3\text{COO})]$

Correct Answer: 5

Solution Analysis: To determine the number of white coloured salts among the listed compounds, we look at their common color characteristics: (a) SrSO_4 - Generally, most sulfate salts are white unless the cation imparts a color. Strontium sulfate is indeed white. (b) $\text{Mg}(\text{NH}_4)\text{PO}_4$ - Magnesium ammonium phosphate is also known to be white. This is typical for many phosphates with colorless cations. (c) BaCrO_4 - Barium chromate is well-known for its yellow color, a characteristic of many chromates. (d) $\text{Mn}(\text{OH})_2$ - Manganese hydroxide can have a faint pink to white color, but in this context, we consider it white. (e) PbSO_4 - Lead sulfate is white. (f) PbCrO_4 - Lead chromate is yellow. (g) AgBr - Silver bromide is white. (h) PbI_2 - Lead iodide is yellow. (i) CaC_2O_4 - Calcium oxalate is white. (j) $[\text{Fe}(\text{OH})_2(\text{CH}_3\text{COO})]$ - Iron(II) acetylacetonate is white. In this context, we consider it white.